

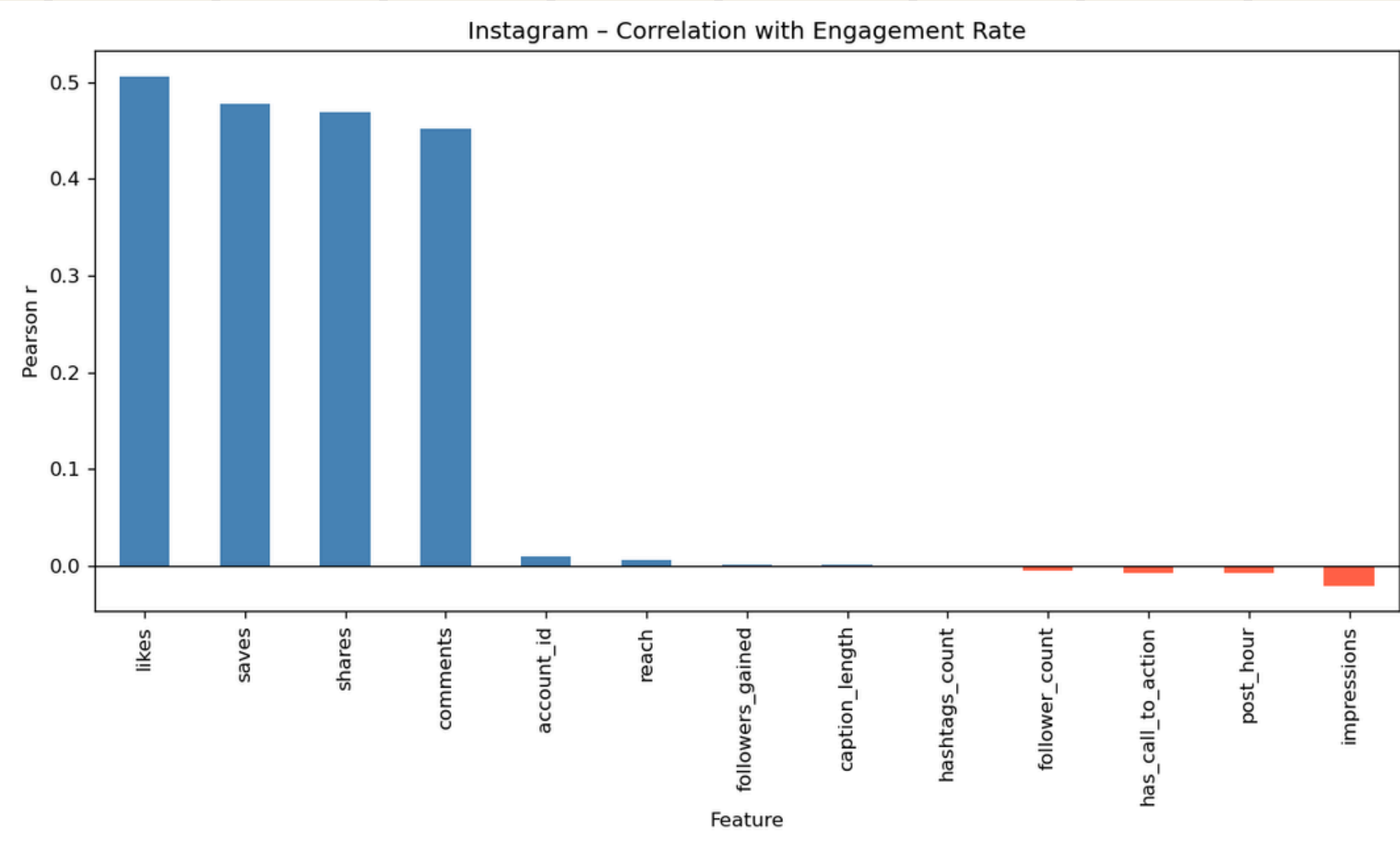
WHAT CONTENT AND TIMING FACTORS DRIVE SOCIAL MEDIA ENGAGEMENT?

Kexuan Liang

Data Source	Name	Link	Format	List of Fileds	Final Data Size
1	Instagram Engagement Analytics	https://www.kaggle.com/datasets/kundanbedml/instagram-engagement-analytics-dataset	CSV	caption_length, hashtags_count, post_hour, likes, comments, engagement_rate	~20,000 posts
2	Viral Social Media Trends	https://www.kaggle.com/datasets/atharvasound/social-media-trends-and-engagement-analysis	CSV	caption_length, hashtags_count, likes, comments, shares, engagement_rate	~5,000 rows
3	YouTube Data API v3	https://developers.google.com/youtube/v3	API	video_id, title, published_at, view_count, like_count, comment_count	~400 videos

Results : Correlation - Instagram

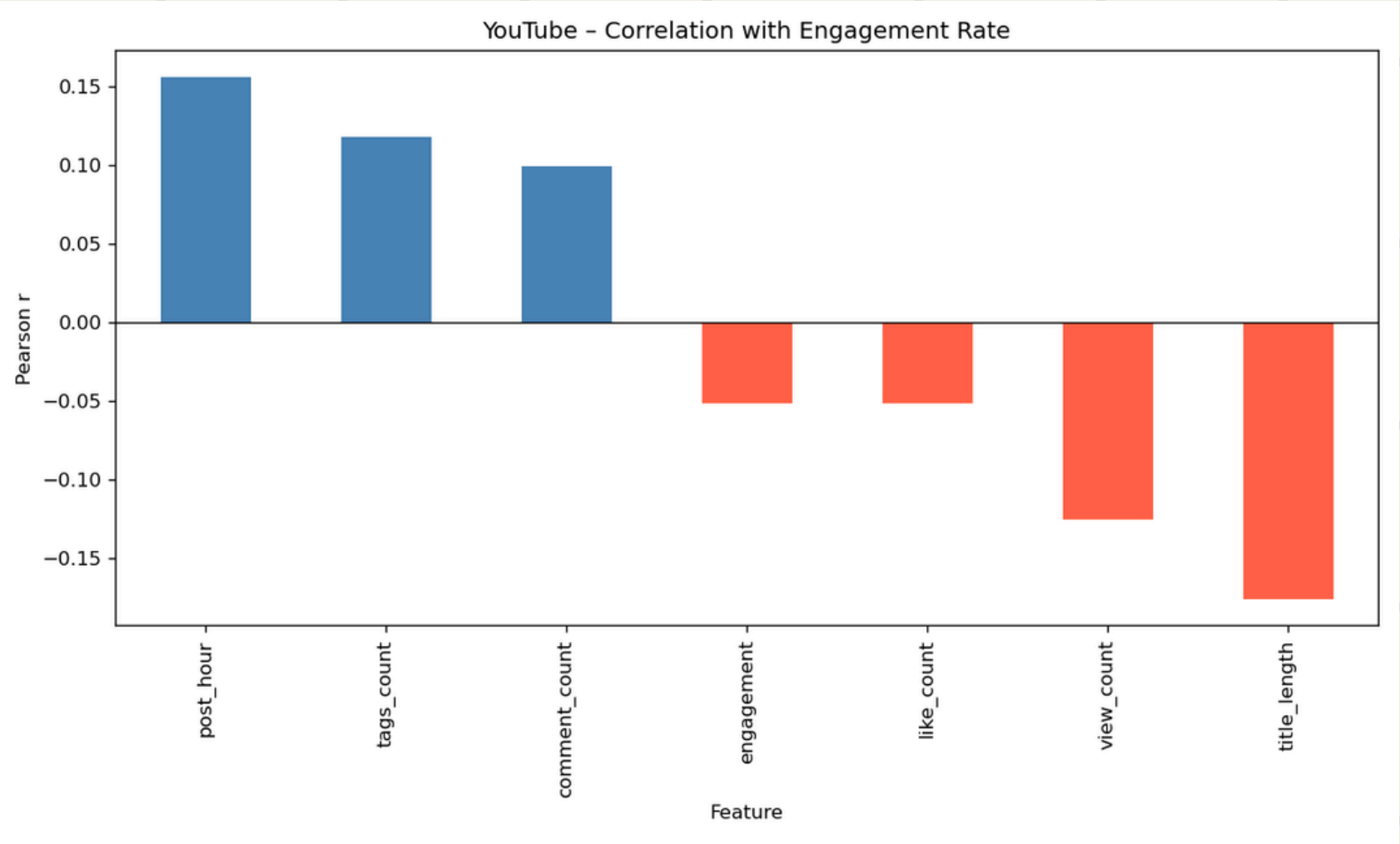
Source: Kaggle Instagram Analytics	
Engagement Rate: Pre-calculated in dataset (exact formula not disclosed)	
~20,000 posts	
FEATURES EXPLAINED	
Feature	What it means
likes	Number of likes on the post
saves	Number of times the post was saved
shares	Number of times the post was shared
comments	Number of comments
impressions	Total number of times the post was seen
caption_length	How long the caption is (number of characters)
hashtags_count	How many hashtags were used
post_hour	What hour of the day the post was published (0-23)
follower_count	Number of followers the account has



- Likes, saves, shares, and comments are all correlated with engagement rate — r values range from 0.45 to 0.51
- Caption length ($r \approx 0.02$) and hashtag count ($r \approx 0.01$) basically don't matter — near-zero correlation with engagement rate
- Impressions are actually negatively correlated ($r \approx -0.05$) — when a post gets boosted or goes to the Explore page, it reaches a lot of people who don't follow you and aren't really interested, so they scroll past without engaging, which drives the engagement rate down
- Takeaway: Engagement in Instagram is driven by audience interest, not content formatting

Results : Correlation - Youtube

Source: YouTube Data API (self-collected)	
Engagement Rate: (likes + comments) ÷ views	
~400 videos	
FEATURES EXPLAINED	
Feature	What it means
title_length	How long the video title is (number of characters)
tags_count	How many tags were added to the video
post_hour	What hour of the day the video was uploaded (0-23)
view_count	Total number of views
like_count	Number of likes
comment_count	Number of comments



- Posting hour ($r \approx 0.15$) number of tags ($r \approx 0.12$), comments ($r \approx 0.10$) have a slight positive correlation with engagement rate
- Title length is negatively correlated ($r \approx -0.18$) — shorter titles perform better
- Takeaway: It's not really about how to format the content — it's more about whether people actually want to interact with it

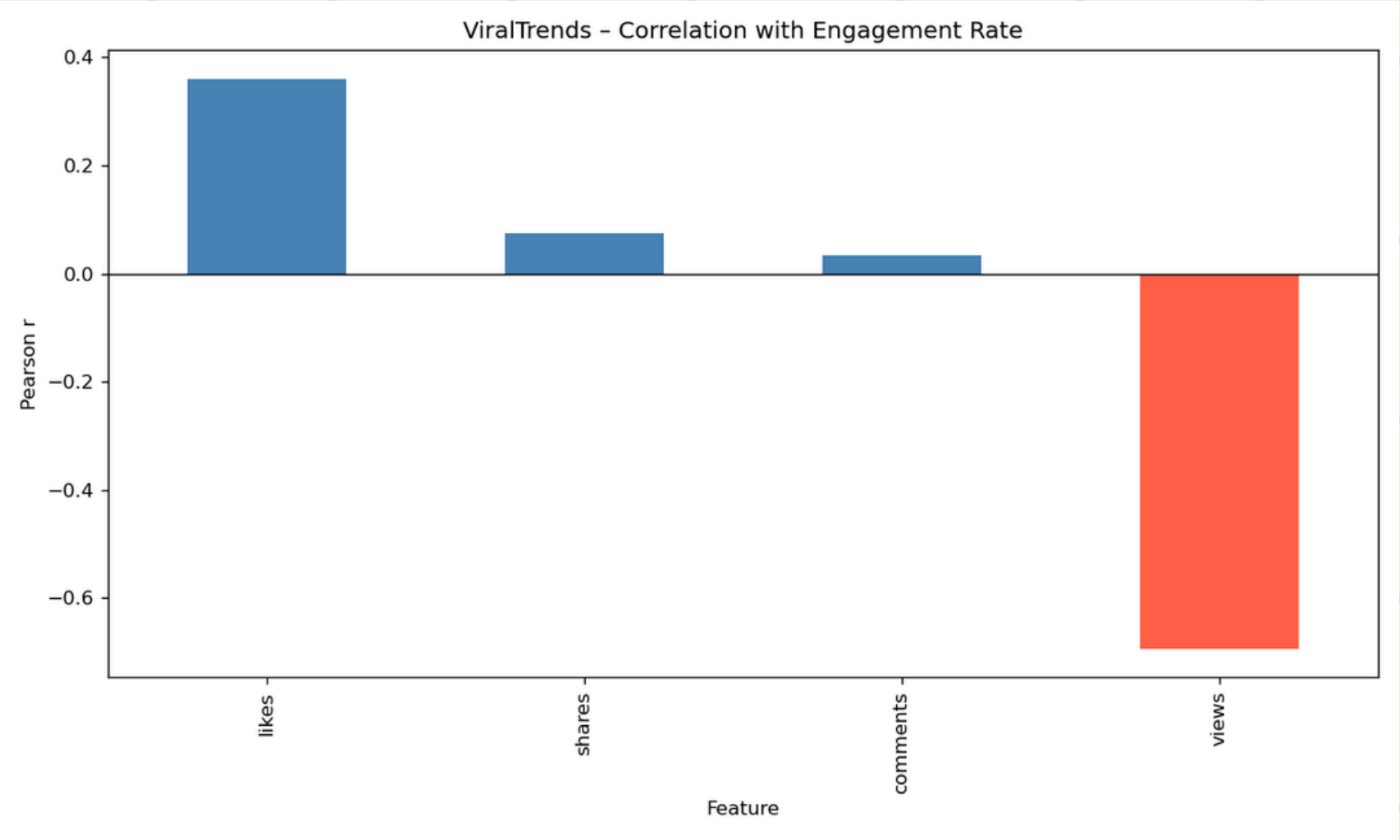
Results : Correlation - Viral Trend

Source: Kaggle Viral Social Media Trends

~5,000 rows

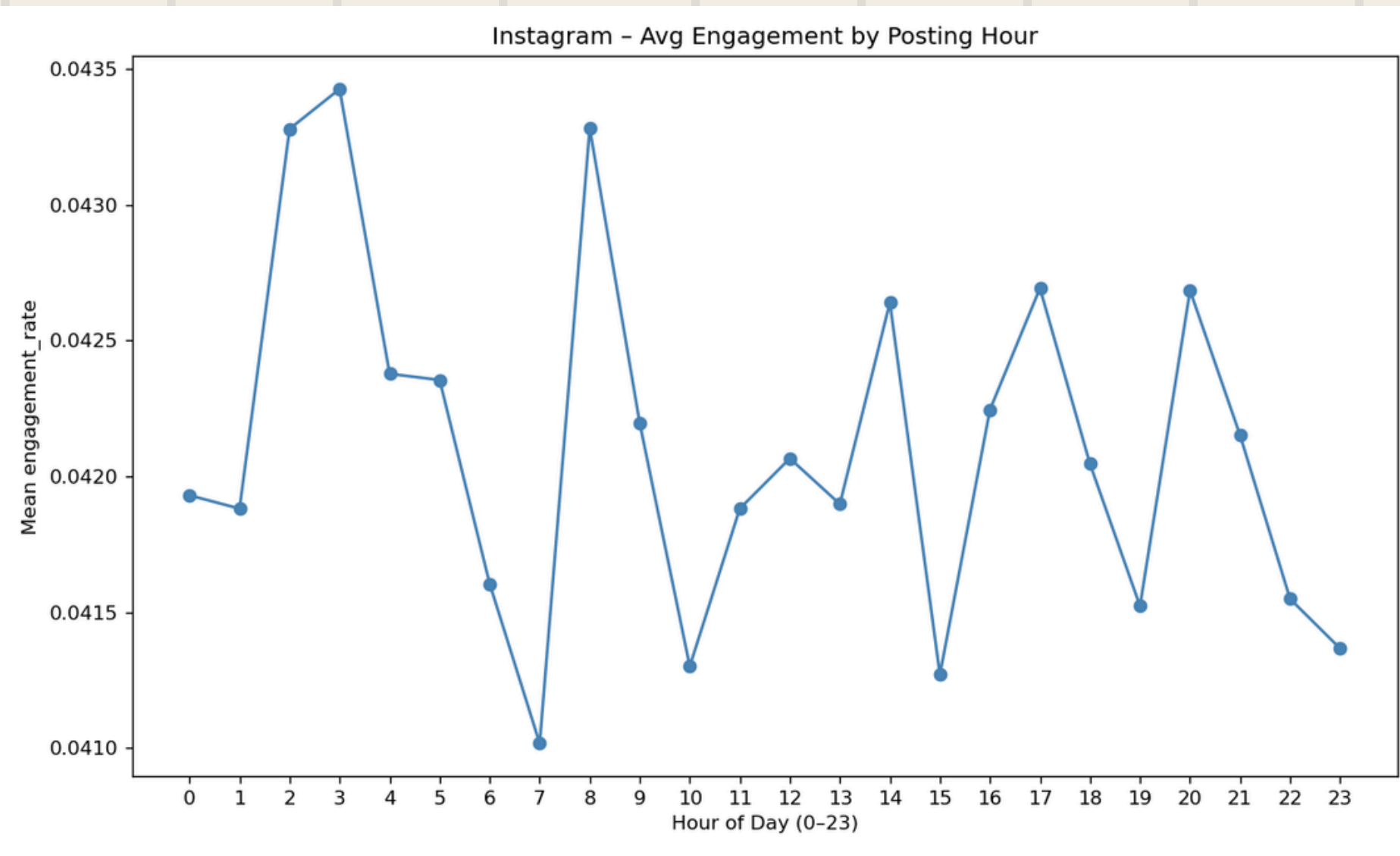
Engagement Rate: (likes + comments + shares) ÷ views

FEATURES EXPLAINED	
Feature	What it means
likes	Number of likes
shares	Number of shares
comments	Number of comments
views	Total number of views
platform	Which platform: Instagram, TikTok, Twitter, YouTube



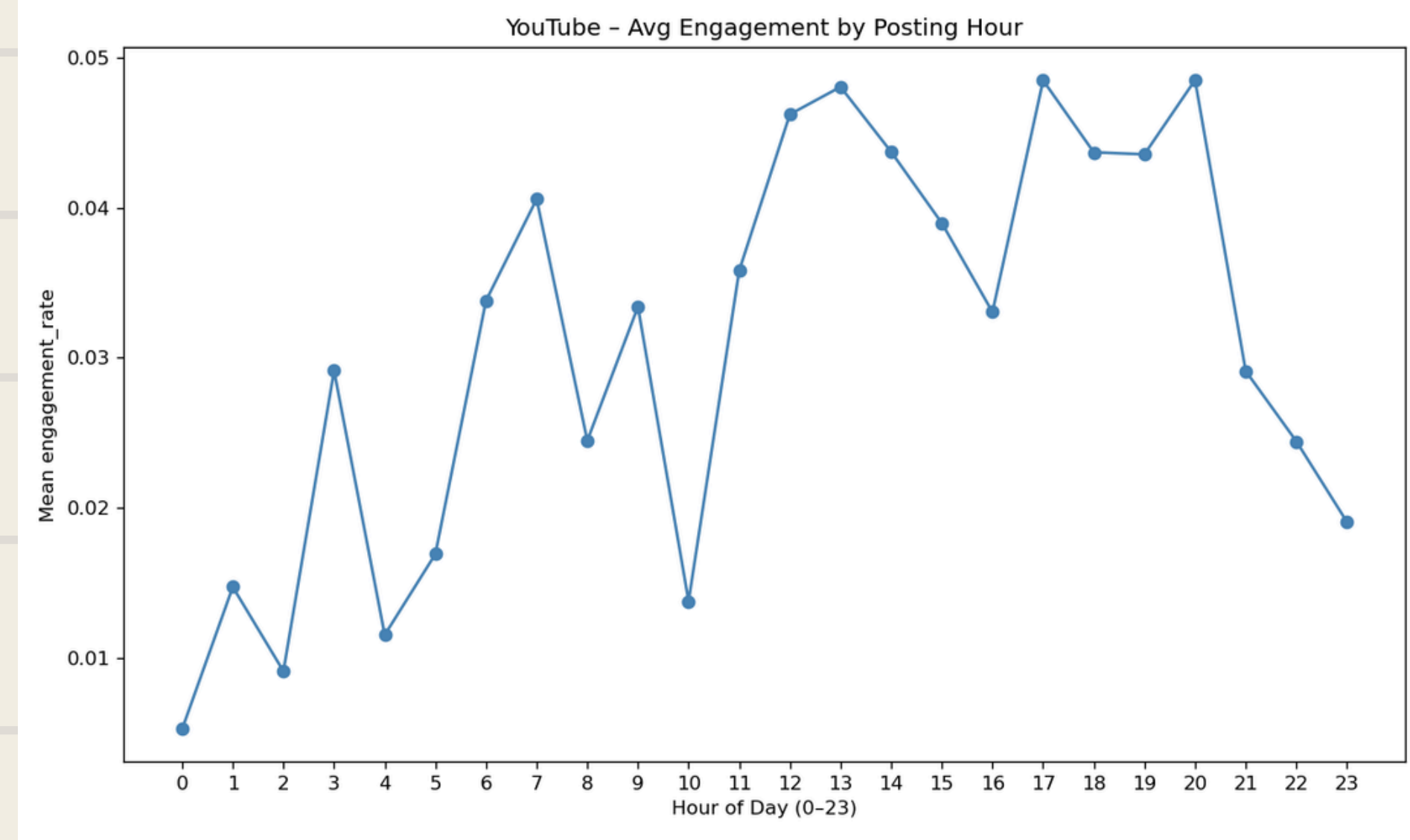
- Likes ($r \approx 0.35$) and shares ($r \approx 0.10$) are positively correlated with engagement rate
- Views are strongly negatively correlated ($r \approx -0.65$) — when content goes viral, the view count (the denominator) grows much bigger, while likes and comments only come from the core audience. So reach is not the same as engagement
- Takeaway: Investing in better content quality and building a loyal audience matters far more than chasing viral reach

Results : Timing



Instagram:

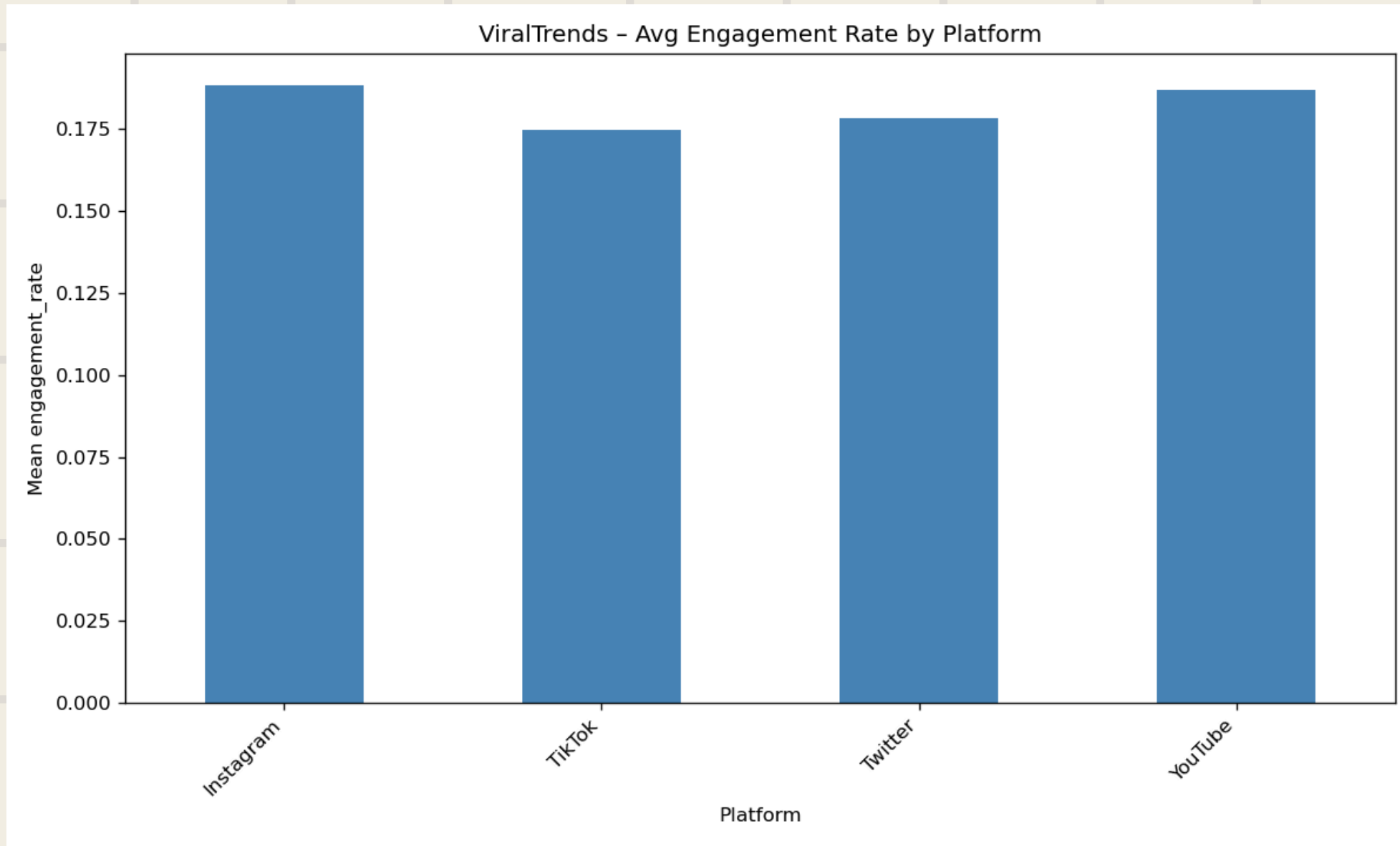
- Engagement rate stays between 0.041 and 0.0434 throughout the day — a range of only 0.0024, which is very small
- Slight peaks around 2–3am and 8am, probably from night owls or people checking their phones first thing in the morning
- Takeaway: timing doesn't really matter much for Instagram



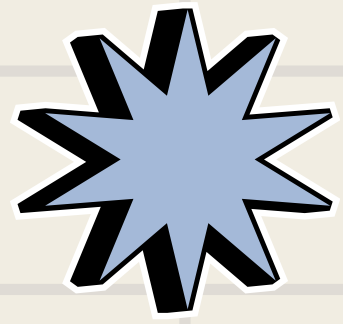
Youtube:

- Engagement starts really low at midnight (around 0.006), then gradually climbs throughout the day, peaking around 1pm, 5pm, and 8pm at about 0.048
- That's a range of over 0.04 — nearly 20 times larger than Instagram's range of 0.0024
- Takeaway: schedule your videos to go live around lunchtime or in the evening when your audience is most likely to be relaxed and ready to watch

Results : Platform



- All four platforms have very similar average engagement rates, between 0.17 and 0.19 — the gap across all platforms is less than 0.02
- Instagram (≈ 0.19) and YouTube (≈ 0.19) are slightly higher; TikTok is the lowest (≈ 0.17); Twitter sits in the middle (≈ 0.18)
- Switching platforms alone won't significantly improve your engagement
- A possible reason: Instagram and YouTube are follow-based — users actively choose to follow accounts and are more likely to see and interact with their content. TikTok is more algorithm-driven, so content is pushed to random users who may be less likely to engage
- Overall takeaway: Across all three datasets, content format factors like caption length and hashtag count have very little impact. What matters more is audience intent and platform behavior. For timing, it depends on the platform — Instagram is flexible, but YouTube rewards afternoon and evening posts



Challenge

- **API Limitation**

- **X API became paid → switched to YouTube Data API v3**

↳ **Designed 8 keyword searches to fit within free quota**

- **Inconsistent Engagement Rate**

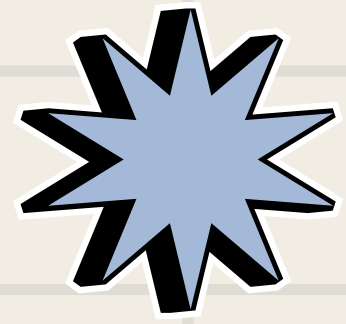
- **Each dataset uses a different formula**

↳ **Analyzed each dataset independently; defined per-dataset formulas; avoided cross-platform comparisons**

- **Missing Posting Hours**

- **Viral Trends only has dates, no specific times**

↳ **Skipped hour analysis for that dataset; code auto-detects missing columns and skips relevant plots**



Future Work

- **Integrate Reddit / TikTok APIs for broader social media coverage**
- **Standardize a unified engagement metric across platforms**
- **Find datasets with both date and time stamps for cross-platform timing analysis**